Informatica® Data Ingestion and Replication
October 2026

Data Ingestion and Replication Command-Line Interface

Informatica Data Ingestion and Replication Data Ingestion and Replication Command-Line Interface
October 2026

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Preface

Use *Data Ingestion and Replication Command-Line Interface* to get information about the CLI commands you can use to create, view, and deploy tasks and to list, manage, and monitor application ingestion and replication and database ingestion and replication jobs.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

IDMC web site

You can access the Informatica® Intelligent Data Management Cloud™ (IDMC) web site at http://www.informatica.com/cloud.

IDMC Communities

Use the IDMC Community to discuss and resolve technical issues. You can also find technical tips, documentation updates, and answers to frequently asked questions.

Access the IDMC Community at:

https://network.informatica.com/community/informatica-network/products/cloud-integration

Developers can learn more and share tips at the Cloud Developer community:

https://network.informatica.com/community/informatica-network/products/cloud-integration/cloud-developers

IDMC Marketplace

Visit the Informatica Marketplace to try and buy Data Integration Connectors, templates, and mapplets:

https://marketplace.informatica.com/

Data Integration connector documentation

You can access documentation for Data Integration Connectors at the Documentation Portal. To explore the Documentation Portal, visit https://docs.informatica.com.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

IDMC Trust Center

The IDMC Trust Center provides information about Informatica security, compliance, and privacy practices.

You can access the trust center at https://trust.informatica.com/.

IDMC Pulse Dashboard

The IDMC Pulse Dashboard provides system status and upgrade information for all PODs and services across IDMC.

Follow the services you use and subscribe to alerts to receive notifications about upgrades, maintenance, production incidents, and upcoming releases. The IDMC Pulse Dashboard displays the production status of all the IDMC products. All maintenance updates are posted to this page, and during an outage, it will have the most current information. To ensure you are notified of updates and outages, you can subscribe to receive updates for a single service or all IDMC services. Subscribing to all services is the best way to be certain you never miss an update.

For more information, see the Service Status and Communications guide.

Informatica Global Customer Support

You can contact a Global Support Center through the Informatica Network or by telephone.

To find online support resources on the Informatica Network, click **Contact Support** in the IDMC Help menu to go to the **Cloud Support** page. The **Cloud Support** page includes system status information and community discussions. Log in to Informatica Network and click **Need Help** to find additional resources and to contact Informatica Global Customer Support through email.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Data Ingestion and Replication Command-Line Interface

You can run commands in the Data Ingestion and Replication Command Line Interface (CLI) to create, view, and deploy application ingestion and replication and database ingestion and replication tasks and to list, start, stop, redeploy, undeploy, and monitor ingestion jobs.

Data Ingestion and Replication CLI overview

The Data Ingestion and Replication Command-Line Interface (CLI) files are installed in the `<Informatica Cloud Secure Agent installation>\apps\dbmicli` directory. The directory contains .jar files, sample and customized YAML input files, default and customized dbmiclienv properties files, and any key-store.txt file that is generated if you enabled encryption.

The CLI requires Java Runtime Environment (JRE) 17 or later. If the Java version on the system PATH is earlier than JRE 17, the CLI attempts to use the JRE bundled with the Secure Agent. If neither option provides JRE 17 or later, the CLI exits with a message indicating that JRE 17 or later is required.

You can use Data Ingestion and Replication CLI commands to perform the following tasks for application ingestion and replication jobs and database ingestion and replication jobs:

- Create an encrypted password that you can use in CLI commands or connection properties in the user interface.
- Create a new task or create tasks based on one or more existing tasks.
- Deploy a task.
- View status information for your organization's jobs or a specific job.
- Start or resume a job.
- Stop or abort a job.
- Replace an existing task definition with an updated version of the task.
- Undeploy a job.
- Redeploy a job.
- Resynchronize the target with the source in a combined load job.
- Monitor a job.
- View statistics for a job or a subtask of a job.

For database ingestion and replication tasks only, you can use the CLI to enable the database CDC option on the source tables.

Commands are also available to get the following information about the Data Ingestion and Replication CLI:

- Help information about CLI commands
- The current version of the CLI

Required user roles and permissions

For application ingestion and replication tasks and database ingestion and replication tasks, you must have certain roles and permissions to perform tasks by using the CLI. Configure user roles and privileges in Administrator.

The following system-defined user roles are typically required:

- To list all of the application ingestion and replication and database ingestion and replication jobs in your organization, make sure you have one of the following system-defined roles or a custom role that has at least the same privileges:
  - Admin
  - Data Integration Task Executor
  - Designer
  - Monitor

  If you do not have any of these roles, you can list only the jobs that you own.
- To start, resume, stop, or abort jobs for an ingestion and replication task, make sure you have one of the following system-defined roles or a custom role that has at least the same privileges:
  - Admin
  - Designer

  For application ingestion and replication and database ingestion and replication tasks, you must also have the execute permission on the task.
- To create ingestion and replication tasks, make sure you have one of the following system-defined roles or a custom role that has at least the same privileges:
  - Admin
  - Designer
- To deploy ingestion and replication tasks, make sure you have one of the following system-defined roles or a custom role that has at least the same privileges:
  - Admin
  - Designer
  - Deployer

  For application ingestion and replication and database ingestion and replication tasks, you must also have the execute permission on the task.

Running the Data Ingestion and Replication CLI

You can run the CLI by using the scripts installed in the `<Informatica Cloud Secure Agent installation>\apps\dbmicli` directory.

Note: Ensure that Java Runtime Environment (JRE) 17 or later is installed on the machine where you run the CLI. If the Java version on the system PATH is earlier than JRE 17, the CLI attempts to use the JRE bundled with the Secure Agent. If neither option provides JRE 17 or later, the CLI exits with a message indicating that JRE 17 or later is required.

To start the CLI, run one of the following scripts:

- On Linux, run dbmicli.sh
- On Windows, run dbmicli.bat

Syntax

dbmicli [*common options*] *command* [*command-specific options*]

Creating an encrypted password

Use the encryptText command to generate an encrypted value for a plain-text password. You can use the encrypted value in place of a plain-text password in CLI commands and in connection properties in the user interface.

Syntax:

```
encryptText --text password --method method
```

The --method argument is required and determines how the encryption key is managed. Specify one of the following methods:

- selfcontained. Embeds the encryption key directly in the encrypted output. The encrypted value is portable and does not require a separate key file. You can use this method when pasting an encrypted password into connection properties in the user interface.
- keystore. Stores the randomly generated encryption key in a key-store.txt file in the directory that contains the dbmicli .jar files. The CLI creates this file if it does not already exist. If the file already exists, the CLI reuses the existing key for all subsequent encryptions rather than generating a new one. To generate a new encryption key for a different password, move the existing keystore file out of the CLI directory before running the encryptText command again.
  The CLI uses the keystore file to decrypt any encrypted password passed to it at runtime, so the keystore file must be present in the CLI directory where the CLI command is executed. If you run CLI commands from the same directory where the password was encrypted, decryption is seamless. If you encrypt a password on one machine and run the CLI commands on a different machine or directory, you must copy the keystore file to that location.

  Because the user interface has no access to the keystore file, you cannot use keystore-encrypted passwords in connection properties in the user interface. Use the selfcontained method for those passwords.

Example:

The following command generates a portable encrypted password using the selfcontained method:

```
dbmicli.bat encryptText --text MyPassword@123 --method selfcontained
```

Output:

The command returns a base64-encoded encrypted value.

Usage considerations:

- You can provide the encrypted value in the -p or --password parameter of other CLI commands, or paste it directly into the corresponding password field in the connection properties in the user interface. When encrypting Oracle connection credentials, such as database password, Reader ASM Password, and TDE Wallet Password, encrypt each credential separately.
- Test Connection doesn't work for connections whose password fields contain an encrypted value created by the CLI. Validate the connection by deploying it and running a job instead of using Test Connection.
- Encrypted values generated by the CLI are only recognized and decrypted by the Data Ingestion and Replication service. Data Integration and other services can't recognize or decrypt these encrypted values. If an Oracle connection is shared with other Intelligent Data Management Cloud services, use a separate, dedicated connection for Database Ingestion and Replication when using CLI-encrypted passwords.
- The selfcontained encryption option is less secure than an encryption solution that requires both the encrypted value and a separate encryption key. The selfcontained solution is sufficient for obfuscation within your organization. Don’t use it to protect values shared outside of your organization.

Data Ingestion and Replication CLI common options

You must provide the Informatica Data Management Cloud user name, password, and pod URL to execute CLI commands. You can provide this information in one or more of the following ways:.

- Specify the values in command-line options for a CLI command. Syntax is:

  ```
  Secure_Agent_installation_directory\apps\dbmicli\dbmicli.bat -P pod_URL -u username -p password CLI_command
  ```

- Read the values from dbmiclienv.properties file. To set up the file, create a copy of dbmiclienv_default.properties file in the dbmicli directory and then rename the copy to dbmiclienv.properties. In the dbmiclienv.properties file, set the following properties as key=value pairs:

  - user=<*user_name*>

  - password=<*password or encrypted password*>

  - pod=<*pod URL*>

  The property values are case-sensitive.

  The dbmiclienv.properties apply to all CLI commands. If you also specify command-line options that provide this information for a specific CLI command, the command-line options take precedence.

- Define the values in the environment variables. Syntax is:

  ```
  [root@root_user_ID dbmicli]# export DBMIPOD=pod_URL
  [root@root_user_ID dbmicli]# export DBMIUSER=username
  [root@root_user_ID dbmicli]# export DBMIPASSWORD=password
  [root@root_user_ID dbmicli]# ./dbmicli.sh CLI_command
  ```

- Enter them manually in interactive mode. To use the interactive mode, enter the username and password after running the following command:

  ```
  Secure_Agent_installation_directory/apps/dbmicli/dbmicli.bat -P pod_URL CLI_command
  ```

The following table describes the common options of the Data Ingestion and Replication CLI:

Common Option	Environment Variable	Description
-u, --user	DBMIUSER	Name of a Data Ingestion and Replication user.
-p, --password	DBMIPASSWORD	Clear-text password or encrypted password to use with the Data Ingestion and Replication user. **Note:** You can create an encrypted password with the encryptText command. See "Creating an encrypted password" on page 8.
-P, --pod	DBMIPOD	Domain name for your Informatica Data Management Cloud Services pod. **Important:** You can get the domain name of your pod from the web address of your POD. From the web address, omit the first part of the domain name (fourth level domain) and the path. For example, if you have the web address `https://usw1.dm1-us.informaticacloud.com/cloudshell/showProducts`, the domain name of your pod is `dm1-us.informaticacloud.com`.

The Data Ingestion and Replication CLI considers the values for the user name, password, and pod URL in the following order:

1. The values specified in the command-line options.
2. The values read from the dbmiclienv.properties file.
3. The corresponding environment variable values, if defined.
4. If none of the above are defined, the CLI switches to interactive mode and prompts the user to enter the values.

Important: If the pod URL is not specified with the -P option and is not defined in the DBMIPOD environment variable, the CLI ends with an error without switching to the interactive mode.

Data Ingestion and Replication CLI commands

The Data Ingestion and Replication CLI commands apply to both Application Ingestion and Replication and Database Ingestion and Replication, unless otherwise indicated.

The following table describes the CLI commands:

Note: In the CLI, the naming convention for job name is <*taskname*> _<*job_instance_number*>, with an underscore (_).This convention is different from that in the IDMC user interface, which uses <*taskname*>-<*job_instance_number*>, with a hyphen (-).

Command	Description
--version, -v	Displays the version of the Data Ingestion and Replication CLI.
help, --help, -h	Displays a list of all of the Data Ingestion and Replication CLI options and commands.
help *command*	Displays a description of the specified command.

Command	Description
encryptText {-t |--text} *password* {-m|--method} *method*	Generates an encrypted value for the specified password text. You can use the encrypted value in place of a plain-text password in other CLI commands and in connection properties in the user interface. Syntax is: `encryptText --text password --method method` The --method argument is required. Specify one of the following methods: - selfcontained. Embeds the encryption key in the encrypted output. The encrypted value is portable and does not require a separate key file. Use this method for connection passwords entered in the user interface. - keystore. Stores the randomly generated encryption key in a key-store.txt file in the same directory as the dbmicli .jar files. Because the user interface cannot access the keystore file, you cannot use keystore-encrypted passwords in connection properties in the user interface. The CLI creates the key-store.txt file if it does not already exist. If the file exists, the CLI reuses the existing key for all subsequent encryptions. To generate a new encryption key for a different password, move the existing keystore file out of the directory before running the command again. The keystore file must be present in the CLI directory where the CLI command is executed. If you encrypt a password on one machine and run the CLI commands on a different machine or directory, you must copy the keystore file to that location
job start *job_name* [--waitForCompletion] [--period=*requestsPeriod*] [--retryCount=*retryCount*]	Starts the job with the specified job name. Optionally monitors the job and resumes the job if it has the Failed status. Syntax is: `job start job_name [--waitForCompletion] [--period=requestsPeriod] [--retryCount=retryCount]` Options are: - waitForCompletion. Waits for a Completed, Aborted, or Stopped status, or resumes the job if it has the Failed status. - period=*requestsPeriod*. The time, in milliseconds, between job status checks. The default value is 5000. - retryCount=*retryCount*. The number of attempts that can be made to resume the job before stopping job monitoring. The default value is 10. For jobs with a job start --waitForCompletion command, the exit codes indicate the current status of the job. To view the exit code for the job, run one of the following commands: - On Windows, run `echo %ERRORLEVEL%`. - On Linux, run `echo $?`. The following exit codes can be returned for a job start --waitForCompletion command to indicate the job status: - `exit code 0` - Completed - `exit code 1` - An exception occurred - `exit code 2` - Failed and there is no retry left - `exit code 3` - Stopped - `exit code 4` - Aborted - `exit code 5` - Undeployed
job resume *job_name*	Resumes the job that has the specified job name.
job stop *job_name*	Stops the job that has the specified job name.
job abort *job_name*	Aborts the job that has the specified job name.

Command	Description

job status
[-f *field_names*]
[--taskType *task_type*]
	Displays information about your organization's jobs if you have the required permissions. If you do not have sufficient privileges, no jobs are listed.
	Syntax is:
	`job status [-f field1,field2,...] [taskType=task_type]`
	Use the -f option to specify the comma-separated list of fields that you want the CLI to return for each job. By default, the CLI returns the following string of values for all jobs:
	`name;status;startTime;duration`
	To customize the fields, enter the -f option with one or more of the following field names:
	- name. The job name.
	- state. The status of the job. For descriptions of status values, see "Monitoring Ingestion Jobs" > "Job Properties" in the Data Ingestion and Replication online help or documentation on the Doc Portal.
	- startTime. The date and time when the job was deployed.
	- endTime. The date and time when the job ended because it completed processing, was stopped, or failed.
	- undeployTime. The date and time when the job was undeployed.
	- duration. The amount of time that the job has run since it was deployed.
	For example, the following command lists the names and statuses of all jobs:
	`C:\DBMICLI>dbmicli.bat -P MyPod --user user_name --password user_password job status -f name,state`
	Use the taskType option to specify the type of ingestion and replication task, which can be either appmi for application ingestion and replication or dbmi for database ingestion and replication. If you omit this option, both task types are included.
	Example output:
	`testGCS_2_1611;Completed`
	`testGCS_0403_1610;Undeployed`
	`testGCS_0403_1607;Undeployed`
	`testGCS_1528;Up and Running`

job status [-f *field_names*] *job_name*	Displays information for a job with a specific job name.
	Syntax is:
	`job status [-f field1,field2,...] job_name`
	You can use the -f option to specify a comma-separated list of fields that you want the CLI to return for the job.
	You must enter the job name. Then enter one or more of the following field names:
	- state. The status of the job. For descriptions of status values, see "Monitoring Ingestion Jobs" > "Job Properties" in the Data Ingestion and Replication online help or documentation on the Doc Portal.
	- startTime. The date and time when the job was deployed.
	- endTime. The date and time when the job ended because it completed processing, was stopped, or failed.
	- undeployTime. The date and time when the job was undeployed.
	- duration. The amount of time that the job has run since it was deployed.
	For example, the following command lists the start time and duration of the job that has the sample_job name:
	`C:\DBMICLI>dbmicli.bat -P MyPod --user user_name --password user_password job status -f startTime,duration sample_job`
	Example output:
	`2021-03-04T05:39:23.000-0800;02:51:32`

Command	Description

job metrics *job_name* [-f *field_names*]	Displays statistics for a job that has the specified job name.
	Syntax is:
	`job metrics job_name [-f field1,field2,field3]`
	You can use the -f option to specify a comma-separated list of fields that you want the CLI to return for the job.
	Valid fields are:
	- recordsRead. The number of records that were read from the source. Displayed by default.
	- recordsWritten. The number of records that were successfully written to the target. Displayed by default.
	- captureProgress. The date and time in the source change stream to which capture processing has progressed, displayed in UTC time. Not valid for initial load jobs. Valid only for database ingestion and replication tasks.
	Example default output showing records read and records written:
	`40;40`

job metrics *job_name* --subtask *subtask_name* [-f *field_names*]	Displays statistics for a specified subtask of the job that has the specified job name.
	Syntax is:
	`job metrics job_name --subtask subtask_name [-f field1,field2,...]`
	Note: The *subtask_name* must be the exact source or target table name, not a partial or substring name.
	You can use the -f option to specify a comma-separated list of fields that you want the CLI to return for the subtask.
	Valid fields are:
	- recordsRead. The number of records that were read from the source. Valid for initial load jobs.
	- recordsWritten. The number of records that were successfully written to the target. Valid for initial load jobs.
	- inserts. The number of inserts that were applied to the target table. Valid for incremental load and combined initial and incremental load jobs.
	- updates. The number of updates that were applied to the target table. Valid for incremental load and combined initial and incremental load jobs.
	- deletes. The number of deletes that were applied to the target table. Valid for incremental load and combined initial and incremental load jobs.
	- lobs. The number of LOBs that were processed at the target table. Valid for incremental load and combined initial and incremental load jobs.
	- unloadCount. The number of records read from source during the unload phase of a combined initial and incremental load job, before the start of incremental change data processing. Valid for combined initial and incremental load jobs.
	Example default output for initial loads:
	`40;40`
	Example default output for incremental loads:
	`4;2;1;0`
	Example default output for combined initial and incremental loads:
	`2;0;1;0;41`

Command	Description
job attach *job_name* [--period=*requestsPeriod*] [--retryCount=*retryCount*]	Monitors the job with the specified job name and resumes the job if it has the Failed status. The cycle ends when one of the following conditions is met: - A Completed, Aborted, or Stopped status is detected for an initial load job. - An Aborted or Stopped status is detected for an incremental load or combined initial and incremental load job. - The current retry count equals or is greater than the specified retryCount. Syntax is: `job attach job_name [--period=requestsPeriod [--retryCount=retryCount]` Options are: - period=*requestsPeriod*. The time, in milliseconds, between job status checks. The default value is 5000. - retryCount=*retryCount*. The number of attempts that can be made to resume the job before stopping job monitoring. The default value is 10. The exit codes indicate the current status of the job. To view the exit code for the job, run one of the following commands: - On Windows, run `echo %ERRORLEVEL%`. - On Linux, run `echo $?`. The following exit codes can be returned for a job attach command to indicate the job status: - `exit code 0` - Completed - `exit code 1` - An exception occurred - `exit code 2` - Failed and there is no retry left - `exit code 3` - Stopped - `exit code 4` - Aborted - `exit code 5` - Undeployed
job undeploy *job_name*	Undeploys an ingestion job that was previously deployed. Before you attempt to undeploy a job, ensure that it is not running. You might need to undeploy a job when you no longer need to run the job, the job is in the Failed state, or you need to change a connection or property in the associated task that cannot be edited until the job is undeployed. After the job is undeployed, you cannot run it again or redeploy it. If you want to run a job for the associated ingestion and replication task again, you must deploy the task again to create a new job instance.
job redeploy *job_name*	Redeploys a job after you've edited available fields in the associated ingestion and replication task so that the new settings can take effect, without first undeploying the job. You can update the task configuration details only in the user interface.
job resumeWithOptions resync *job_name*	For combined load jobs that have Failed or Running With Warnings state due to a schema drift error, resynchronizes the tables that are in error. You can't use this command if the job uses a source that allows selecting individual columns for replication or if the source does not support schema drift. **Note:** This command is not valid if the job is in the Failed state because the task deployment failed.

Command	Description
job resumeWithOptions resyncRefresh *job_name*	For combined load jobs that are in the Stopped, Aborted, or Failed state and have a source that supports selecting individual columns for data replication, resynchronizes the target tables with the latest source table definitions, including any DDL changes. After the target tables are refreshed, the structure of source and target tables match. This command mimics the behavior of the resumeWithOptions resync command. This command applies only to jobs with sources that handle schema drift. **Note:** This command is not valid if the job is in the Failed state because the task deployment failed.
job resumeWithOptions resyncRetain *job_name*	For combined load jobs that are in the Stopped, Aborted, or Failed state and have a source that supports selecting individual columns for data replication, resynchronizes the same columns that have been processed for CDC, retaining the current structure of the source and target tables. No checks for changes to the source or target table definitions are performed. If source DDL changes affected the source table structure, those changes are not processed. This command applies only to jobs with sources that handle schema drift. **Note:** This command is not valid if the job is in the Failed state because the task deployment failed.
job resumeWithOptions ignore *job_name*	For incremental and combined load jobs that are in the Stopped, Aborted, or Failed state due to a schema drift error, does not replicate DDL changes that occur on the source to the target. This override applies only to tables in an Error state caused by the **Stop Table** or **Stop Job** schema drift settings. This command applies only to jobs with sources that handle schema drift.
job resumeWithOptions stopTable *job_name*	For incremental and combined load jobs that are in the Stopped, Aborted, or Failed state due to a schema drift error, stops processing the source object on which the DDL change occurred. This override applies only to tables in an Error state caused by the configured **Stop Table** or **Stop Job** schema drift settings. This command applies only to jobs with sources that handle schema drift. **Important:** After processing stops, the job cannot retrieve any subsequent data changes from that source object, which might result in data loss on the target. To avoid data loss, resynchronize the affected source and target objects by using the job resumeWithOptions replicate job_name command.
job resumeWithOptions replicate *job_name*	For incremental and combined load jobs that are in the Stopped, Aborted, or Failed state due to a schema drift error, replicates the DDL changes to the target. This override applies only to tables in an Error state caused by the configured **Stop Table** or **Stop Job** schema drift settings. This command applies only to jobs with sources that handle schema drift. **Important:** If you specify the **Replicate** option for Rename Column operations on Microsoft Azure Synapse Analytics targets, the job fails with an error.

Command	Description
job resyncSubtask resync *job_name* --srcSchemaName *source_schema_name* --srcTableName *source_table_name*	For a subtask that is part of a running combined initial and incremental load job, resynchronizes the target table with the latest source table definition, including any DDL changes. The subtask must be in a state other than Queued or Starting. You can't use this command if the job uses a source that allows selecting individual columns for replication or if the source does not support schema drift. Syntax is: `job resyncSubtask resync <job_name> --srcSchemaName <source_schema_name> --srcTableName <source_table_name>` Options are: - srcSchemaName. Required. Source schema name. - srcTableName. Required. Source table name. For application ingestion and replication tasks, use the following values as the *source_schema_name*, depending on the source connector used: - For Marketo V3: *MARKETO* - For Microsoft Dynamics 365 Mass Ingestion: *MSD365* - For Netsuite Mass Ingestion: *NETSUITE* - For Salesforce Mass Ingestion: *SALESFORCE* - For Salesforce Marketing Cloud: *SALESFORCE_MARKETING_CLOUD* - For SAP Mass Ingestion: *SAP_ABAP* - For SAP ODP Extractor: *<Context_Name>* - For ServiceNow Mass Ingestion: *SERVICENOW* - For Workday Mass Ingestion: *<Product_Name>*
job resyncSubtask resyncRefresh *job_name* --srcSchemaName *source_schema_name* --srcTableName *source_table_name*	For a subtask in a running combined initial and incremental load job with a source that supports selecting individual columns for data replication, resynchronizes the target table with the latest source table definition, including any DDL changes. The subtask must be in a state other than Queued or Starting. After the target table is refreshed, the target table structure matches the current source table structure. This command mimics the behavior of the resyncSubtask resync command. Syntax is: `job resyncSubtask resyncRefresh <job_name> --srcSchemaName <source_schema_name> --srcTableName <source_table_name>` Options are: - srcSchemaName. Required. Source schema name. - srcTableName. Required. Source table name. **Note:** For SQL Server sources, the *source_table_name* must be in *schema_name.table_name* format. For application ingestion and replication tasks, use the following values as the *source_schema_name*, depending on the source connector used: - For Marketo V3: *MARKETO* - For Microsoft Dynamics 365 Mass Ingestion: *MSD365* - For Netsuite Mass Ingestion: *NETSUITE* - For Salesforce Mass Ingestion: *SALESFORCE* - For Salesforce Marketing Cloud: *SALESFORCE_MARKETING_CLOUD* - For SAP Mass Ingestion: *SAP_ABAP* - For SAP ODP Extractor: *<Context_Name>* - For ServiceNow Mass Ingestion: *SERVICENOW* - For Workday Mass Ingestion: *<Product_Name>*

Command	Description
job resyncSubtask resyncRetain *job_name* --srcSchemaName *source_schema_name* --srcTableName *source_table_name*	For a subtask in a running combined initial and incremental load job with a source that supports selecting individual columns for data replication, resynchronizes the same columns that have been processed for CDC, retaining the current structure of the source and target tables. The subtask must be in a state other than Queued or Starting. No checks for changes in the source or target table definitions are performed. If source DDL changes affected the source table structure, those changes are not processed. Syntax is: `job resyncSubtask resyncRetain <job_name>job_name> --srcSchemaName <source_schema_name> --srcTableName <source_table_name>` Options are: - srcSchemaName. Required. Source schema name. - srcTableName. Required. Source table name. **Note:** For SQL Server sources, the *source_table_name* must be in *schema_name*.*table_name* format. For application ingestion and replication tasks, use the following values as the *source_schema_name*, depending on the source connector used: - For Marketo V3: *MARKETO* - For Microsoft Dynamics 365 Mass Ingestion: *MSD365* - For Netsuite Mass Ingestion: *NETSUITE* - For Salesforce Mass Ingestion: *SALESFORCE* - For Salesforce Marketing Cloud: *SALESFORCE_MARKETING_CLOUD* - For SAP Mass Ingestion: *SAP_ABAP* - For SAP ODP Extractor: *<Context_Name>* - For ServiceNow Mass Ingestion: *SERVICENOW* - For Workday Mass Ingestion: *<Product_Name>*

Command	Description

task create
--pathToConfig *path_to_configuration_file*
[--taskName *task_name*]
[--taskLocation *task_location*]
[--taskType *task_type*]
[--deploy]
[--sourceConnectionId *source_connection_id*]
[--targetConnectionId *target_connection_id*]
[--runtimeEnvId *runtime_environment_id*]
	Creates an ingestion and replication task by using the information from the input configuration file and optionally deploys the task.
	Syntax is:
	`task create --pathToConfig path_to_configuration_file [--taskName task_name][--taskLocation task_location][--taskType task_type][--deploy][--sourceConnectionId source_connection_id][--targetConnectionId target_connection_id][--runtimeEnvId runtime_environment_id]`
	Options are:
	- pathToConfig. Required. The path to the YAML input file.
	- taskName. The task name.
	- taskLocation. The project or project\folder that will contain the task definition. The default value is Default.
	- taskType. The type of task, which can be either appmi for application ingestion and replication or dbmi for database ingestion and replication.
	If you do not specify the task type in the command, you can include it at the beginning of the YAML file.
	Default is dbmi.
	- deploy. Deploys the task after it is created.
	- sourceConnectionId. Overrides the source connection ID in the configuration file. The configuration file is not modified.
	- targetConnectionId. Overrides the target connection ID in the configuration file. The configuration file is not modified.
	- runtimeEnvId. Overrides the runtime environment ID in the configuration file. The configuration file is not modified.
	When you specify --sourceConnectionId, --targetConnectionId, or --runtimeEnvId, the command-line values take precedence over the values in the configuration file. You can specify any combination of the three options. Options that you omit retain the values from the configuration file. The CLI displays a message for each override before executing the command. Use these options to promote a task configuration from one environment to another without editing the file.
	For example, the following command creates a task by using the sample_task.yaml file and overrides the connection and runtime environment IDs from the configuration file:
	`C:\DBMICLI>dbmicli.bat -P MyPod --user user_name --password user_password task create --pathToConfig C:\DBMI_TASKS\sample_task.yaml --sourceConnectionId source_connection_id --targetConnectionId target_connection_id --runtimeEnvId runtime_environment_id`

Command	Description
task createFrom --taskName *task_name* [--taskLocation *task_location*] [-taskType *task_type*] --override *path_to_override_file* [--deploy]	Creates a task from another specified task using the field overrides in an override file. Also optionally deploys the task. For example, if the override file specifies the location "project1" to override the Default location, the new task definition will be in the project1 location. Syntax is: `task createFrom --taskName task_name [--taskLocation task_location][--taskType task_type] --override path_to_override_file [--deploy]` Options are: - taskName. Required. The task name. - taskLocation. The project or project\folder that will contain the task definition. The default value is Default. - taskType. The type of task, which can be either appmi for application ingestion and replication or dbmi for database ingestion and replication. If you do not specify the task type in the command, you can include it at the beginning of the YAML file. Default is dbmi. - override. Required. The path to the YAML file that contains the fields that are used to override fields of the original task. - deploy. Deploys the tasks after they are created.

Command	Description
task multipleCreateFrom --taskName *taskName* --override *path_to_override_file* [--taskLocation *task_location*] [--taskType *task_type*] [--deploy] [--deployTimeout *wait_time_for_status*] [--continueOnFail]	Creates multiple tasks from another task using the field overrides specified in the override file. Also optionally deploys the tasks. Syntax is: `task multipleCreateFrom --taskName task_name --override path_to_override_file [--deploy] [--deployTimeout wait_time_for_status] [--continueOnFail][--taskLocation task_location] [--taskLocation task_location] [--taskType task_type]` Options are: - taskName. Required. The name of task from which you're creating other tasks. - override. Required. The YAML file that contains the field overrides. - taskLocation. The project or project\folder that will contain the new task definitions. The default value is Default. - taskType. The type of task, which can be either appmi for application ingestion and replication or dbmi for database ingestion and replication. If you do not specify the task type in the command, you can include it at the beginning of the YAML file. Default is dbmi. - deploy. Deploys the new tasks after they are created. - deployTimeout. The time interval, in seconds, that Data Ingestion and Replication waits for the deployment status, which can be either deployed or failed, before timing out. You can use this option when you are creating and deploying more than 100 tasks to help keep the Secure Agent running. If a failure occurs, a message is written in the output and to the error log to indicate the status of the job. For example: `Failed;Mass Ingestion could not deploy the task because supplemental logging is not enabled for source table 'TEST169'.` Valid values are 0 to 86400. If you do not specify this option when you are creating and deploying many tasks, requests are sent asynchronously and do not wait for a response, which can lead to increased resource consumption. - continueOnFail. If the creation of a task fails, continues creating the other tasks. For example, if you create 10 tasks and the second task fails, the other tasks will be created.

Command	Description
task cdc --taskName=*taskName* [--taskLocation=*Default*] --scope=*pk*|*all* [--execute] [--download] [--path=*path*] [--msRoleName=*role*] [--msFileGroupName=*group*]	For database ingestion and replication tasks, enables the database CDC option on the source tables. You can optionally generate a CDC script and then either run the script or download it and save it in a file. Syntax is: `task cdc --taskName=taskName[--taskLocation=Default] --scope=pk|all [--execute] [--download][ --path][--msRoleName=role][ --msFileGroupName=group]` Options are: - taskName. Required. The task name. - taskLocation. The project or project\folder that will contain the task definition. The default value is Default. - scope. Required. Generate or run a CDC script for either primary key (PK) columns or for all (ALL) columns. - execute. Execute the CDC script. - download. Download the CDC script. - path. An optional path to the CDC script location. You can specify a file or folder. For example, to specify a script file name and location, enter `C:\cli\cdc.txt`. The CLI then saves the CDC script to the cdc.txt file in the cli directory. To specify a folder only, enter `C:\cli\`. The CLI then saves the generated CDC script file to the cli folder. If a path value is not specified, the script file is saved to the CLI location and file: *cliLocation*\cdc_script_*taskName_currentTimestamp*.txt The following options are valid for Microsoft SQL Server database ingestion and replication sources only: - msRoleName. The name of the database role that is used to gate access to change data. If you do not specify a role name, the database does not use the gating role. - msFileGroupName. The name of the filegroup to be used for the change table that is created for the capture. If you do not specify a filegroup name, the change table is located in the default filegroup if the database.
task deploy [--taskName *taskName*] [--taskLocation *task_location*]	Deploys an application ingestion and replication or database ingestion and replication task. Syntax is: `task deploy [--taskName taskName] [--taskLocation task_location]` Options are: - taskName. The task name. - taskLocation. The project or project\folder that contains the task definition.

Command	Description

task replace
--pathToConfig *yaml_file*
[--taskName *task_name*]
[--taskLocation *task_location*]
[--deploy]
[--noprompt]
[--sourceConnectionId *source_connection_id*]
[--targetConnectionId *target_connection_id*]
[--runtimeEnvId *runtime_environment_id*]
[--retainExistingIds]
	Replaces a task definition with an updated version of the task. You can only replace undeployed tasks or tasks that have never been deployed. Optionally, you can include the --deploy option to also deploy the updated task.
	Syntax is:
	`task replace --pathToConfig path_to_configuration_file [--taskName task_name][--taskLocation task_location][--deploy][--noprompt][--sourceConnectionId source_connection_id][--targetConnectionId target_connection_id][--runtimeEnvId runtime_environment_id][--retainExistingIds]`
	Options are:
	- pathToConfig. Required. Points to the CLI YAML configuration file that you updated or created.
	- taskName. The task name.
	- taskLocation. The project or project\folder that will contain the task definition.
	- deploy. Deploys the updated task definition.
	- noprompt. Causes the following confirmation prompt to not be displayed for a replace operation so that automated scripts and tasks can proceed:
	`The "Replace" command overwrites the current task definition with the given configuration file. Type "Y" to confirm your request:`
	- sourceConnectionId. Overrides the source connection ID in the configuration file. The configuration file is not modified.
	- targetConnectionId. Overrides the target connection ID in the configuration file. The configuration file is not modified.
	- runtimeEnvId. Overrides the runtime environment ID in the configuration file. The configuration file is not modified.
	- retainExistingIds. Retains the sourceConnectionId, targetConnectionId, and runtimeEnvId from the existing task in the repository and ignores values in the configuration file. Takes precedence over --sourceConnectionId, --targetConnectionId, and --runtimeEnvId if those are also supplied.
	Note: If the task does not already exist in the repository, the following error message is displayed:
	`Task config not found for task with name <name> and location <location>`
	When multiple override mechanisms are used together, Data Ingestion and Replication applies them in the following order of precedence:
	1. --retainExistingIds
	2. --sourceConnectionId, --targetConnectionId, --runtimeEnvId
	3. Values in the configuration file.
	If a CLI YAML configuration file already exists, you can modify it. If you created the task from the task wizard and a CLI configuration file does not exist, you can create the YAML configuration file by using the example template file provided in the DBMI package in the following directory:
	`<Secure Agent installation directory>\apps\dbmicli\examples`
	Note: The command does not validate or compare the existing task with the updated configuration file during execution. Make sure the replacement configuration file is correct.
	When you specify --sourceConnectionId, --targetConnectionId, or --runtimeEnvId, the command-line values take precedence over the values in the configuration file. You can specify any combination of the three options. Options that you omit retain the values from the configuration file. The CLI displays a message for each override before executing the command. Use these options to promote a task configuration from one environment to another without editing the file.

Command	Description

	For example, the following command replaces a task by using the sample_task.yaml file and overrides the connection and runtime environment IDs from the configuration file:
	`C:\DBMICLI>dbmicli.bat -P MyPod --user user_name --password user_password task replace --pathToConfig C:\DBMI_TASKS \sample_task.yaml --sourceConnectionId source_connection_id --targetConnectionId target_connection_id --runtimeEnvId runtime_environment_id`
	The following command replaces a task by using the sample_task.yaml file and retains the connection and runtime environment IDs from the existing task in the repository:
	`C:\DBMICLI>dbmicli.bat -P MyPod --user user_name --password user_password task replace --pathToConfig C:\DBMI_TASKS \sample_task.yaml --retainExistingIds`

task status
[-f *field_names*]
[--taskName *taskName*]
[--taskLocation *task_location*]
[--taskType *task_type*]
	Displays information about ingestion and replication tasks.
	Syntax is:
	`task status [-f field1,field2,...] [--taskName taskName] [--taskLocation task_location] [--taskType task_type]`
	By default, the CLI returns the following string of values for each task:
	`name;loadType;sourceConnection;targetConnection;jobName;jobStatus`
	To modify the fields for which values are returned, use the -f option with one or more of the following field names:
	- name. The task name.
	- location. The project or project\folder that will contain the task definition.
	- runtimeEnvironment. The runtime environment in which the task will run.
	- loadType. The type of load operation that the database ingestion and replication task performs.
	- sourceConnection. The connection for the source system.
	- sourceSchema. The source schema that includes the source tables or objects.
	- targetConnection. The connection for the target type.
	- targetSchema. The target schema in which the target tables are created.
	- schedule. The name of a predefined schedule that you want to use for running job instances for an initial load task.
	- jobName. The job name.
	- jobStatus. The job status.
	To filter the tasks, include one or more of the following options:
	- taskName. The name of the task for which to display information.
	- taskLocation. The project or project\folder that contains the task definition.
	- taskType. The type of task, which can be either appmi for application ingestion and replication or dbmi for database ingestion and replication. If you do not specify this option, both ingestion types are included.

Input file format

The Data Ingestion and Replication Command-Line Interface uses input files that have the YAML format to create application ingestion and replication or database ingestion and replication tasks. The YAML (.yml) files are validated for possible errors such as missing required fields or incorrect field names. Examples

of .yml files are available under the dbmicli/examples directory, as a part of secure agent files. Copy an example file under another name and then customize it, as needed.

Tip: Informatica recommends that you copy the dbmicli directory that is installed under the Informatica Cloud Secure Agent to a remote location and then add your YAML files in a folder there. This practice prevents subsequent Secure Agent upgrades from overwriting your YAML files. When you issue commands such as task create, you can specify the path to the YAML file that you want to use.

Initial parameter

You can enter the following parameters at the beginning of the YAML file, before the General section.

taskType

The type of task, which can be either "appmi" for application ingestion and replication tasks or "dbmi" for database ingestion and replication tasks.

If you do not specify the task type in the input file, you can include it in the task create, task createFrom, or task multipleCreateFrom command.

Default is dbmi.

General section parameters

The general section of the YAML input file specifies some basic information about the task, such as a task name, project or project folder location, and load operation type.

name

A name for the ingestion and replication task.

description

An optional description for the task.

location

The project or project\folder that will contain the task definition.

runtimeEnvironment

The runtime environment in which you want to run the task.

type

The type of load operation that the application ingestion and replication or database ingestion and replication task performs. Valid values are:

- initial
- incremental
- combined (for initial and incremental)

Source section parameters

The source section of the YAML input file specifies the source options. Unless otherwise indicated, the options pertain to both application ingestion and replication and database ingestion and replication.

connection

Sources: All

The name of the connection for the source system.

salesforceAPI

Sources: Salesforce, initial load tasks and combined initial and incremental load tasks.

For application ingestion and replication tasks only, the type of Salesforce API that you want to use to retrieve the source data. Valid values are:

- **Standard (REST) API**: Replicates source fields of Base64 data type.
- **Bulk API 2.0**: Excludes replication of source fields of Base64 data type. By default, Bulk API 2.0 is set as the API type in the YAML file.

schema

Sources: All database ingestion and replication sources except MongoDB

For database ingestion and replication tasks only, the source schema that includes the source tables.

database

Sources: MongoDB

For database ingestion and replication tasks only, the MongoDB database that stores the collections with the source data.

journalName

Sources: Db2 for i, incremental loads

For database ingestion and replication tasks only, the name of the journal that records the changes made to the source tables.

replicationSlotName

Sources: PostgreSQL

For database ingestion and replication tasks only, the unique name of a PostgreSQL replication slot.

replicationPlugin

Sources: PostgreSQL

For database ingestion and replication tasks only, a PostgreSQL replication plugin. Valid values are:

- pgoutput
- wal2json

publication

Sources: PostgreSQL

For database ingestion and replication tasks only, the publication name that is used by the pgoutput plugin. Use this parameter only if you specified pgoutput as the replicationPlugin value.

selectionRules

Sources: All

Optional object or table selection rules to select a subset of the source objects or tables. By default, an Include rule that contains only the asterisk (*) wildcard character is provided. This rule selects all objects on the source for an application ingestion and replication task or all tables in the source schema for a database ingestion and replication task. To narrow the source objects or tables to be processed by the task, you can define additional Include rules, Exclude rules, or both types of rules.

Example:

```
selectionRules:
    - include: TABLE_1
    - exclude: TABLE_2
```

restartPointForIncrementalLoad

Sources: SAP ECC sources of application ingestion incremental load jobs. All sources of database ingestion incremental load and combined initial and incremental load jobs.

The position in the source change stream or logs from which an application ingestion and replication or database ingestion and replication job starts reading change records the first time it runs. Valid types are:

- earliest
- latest

For example:

```
restartPointForIncrementalLoad: earliest
restartPointForIncrementalLoad: latest
```

For database ingestion and replication jobs, the default is latest.

For application ingestion and replication jobs, the **earliest** option is not supported and the **latest** option applies only to SAP ECC sources. The default is the restartPointForIncrementalLoadTimestamp value.

If you specify this parameter, do not also specify the restarPointForIncrementalLoadPosition or restartPointForIncrementalLoadTimestamp parameter.

restartPointForIncrementalLoadPosition

Sources: All sources of database ingestion incremental load and combined initial and incremental load jobs.

The RBA position in the source change stream or logs from which a database ingestion and replication job starts reading change records the first time it runs. For example:

```
restartPointForIncrementalLoadPosition: 0
```

This parameter is not supported for application ingestion jobs.

If you specify this parameter, do not also specify the restarPointForIncrementalLoad or restartPointForIncrementalLoadTimestamp parameter.

restartPointForIncrementalLoadTimestamp

Sources: All sources of application incremental load jobs and of database ingestion incremental load and combined initial and incremental load jobs.

The date and time, including AM or PM, in the source change stream or logs from which a application ingestion and replication or database ingestion and replication job starts reading change records the first time it runs. For example:

```
restartPointForIncrementalLoadTimestamp: 2021-08-18 02:50:00 PM
```

For application ingestion jobs, this parameter determines the default behavior.

If you specify this parameter, do not also specify the restarPointForIncrementalLoad or restartPointForIncrementalLoadPosition parameter.

cdcinterval

Sources: Salesforce, incremental load tasks and combined initial and incremental load tasks

For application ingestion and replication tasks only, the time interval in which the application ingestion and replication job runs to retrieve change records for incremental load. You can configure the cdcIntervalDays, cdcIntervalHours, and cdcIntervalMins parameters to specify the interval.

fetchSize

Sources: Salesforce, initial load tasks and incremental load tasks

For application ingestion and replication tasks only, the number of records that the application ingestion and replication job reads at a time from the source. The default value for initial load tasks is 50000 and the default value for incremental load tasks is 2000. If you specified Standard (REST) API for the salesforceAPI parameter, you must change the fetchSize value to 2000.

fetchSizeForInitialLoad

Sources: Salesforce, combined initial and incremental load tasks

For application ingestion and replication tasks only, the number of source records that the application ingestion and replication job reads at a time during the initial loading of data. The default value is 50000. If you specified Standard (REST) API for the salesforceAPI parameter, you must change the fetchSize value to 2000.

fetchSizeForIncrementalLoad

Sources: Salesforce, combined initial and incremental load tasks

For application ingestion and replication tasks only, the number of source records that the application ingestion and replication job reads at a time during the incremental loading of change data. The default value is 2000. For initial load tasks or combined initial and incremental load tasks, if you specified Standard (REST) API for the salesforceAPI parameter, you must change the fetchSize value to 2000.

includeArchivedAndDeletedRows

Sources: Salesforce, initial load tasks and combined initial and incremental load tasks

For application ingestion and replication tasks only, ingests archived and soft-deleted rows from the source during the initial loading of data. The default value is false.

includeBase64Fields

Sources: Salesforce

For application ingestion and replication tasks only, ingests the source fields of Base64 data type. The default value is false.

Note: Configure this parameter only if you set the salesforceAPI parameter to Standard (REST) API .

maximumBase64BodySize

Sources: Salesforce

For application ingestion and replication tasks only, the maximum body size of Base64 encoded data. The default value is 7 MB.

Note: Configure this parameter only if you set the includeBase64Fields parameter to yes.

includeViews

Sources: Microsoft SQL Server, Oracle

For database ingestion and replication tasks only, indicates whether to include views in the table counts and list of table names for database ingestion and replication tasks. Valid values:

- true. Include views.
- false. Do not include views.

customProperties

Sources: All

Custom properties that Informatica provides to meet special requirements. Specify these properties only at the direction of Informatica Global Customer Support.

Example:

```
customProperties:
    "readerInputIsPersisted": true
    "propertyName": "propertyValue"
```

Target section parameters

The target section of the YAML input file specifies the target options.

connection

Targets: All

The name of the connection for the target system.

schema

Targets: Amazon Redshift, Databricks, Google BigQuery, Microsoft Azure Synapse Analytics, Oracle, Snowflake

The target schema in which Data Ingestion and Replication creates the target objects or tables.

bucket

Targets: Amazon Redshift, Google Cloud Storage, Google BigQuery

Specifies the name of an existing bucket container that stores, organizes, and controls access to the data objects that you load to the target.

directory

Targets: Databricks, Amazon Redshift, Google Cloud Storage, Google BigQuery

Specifies the virtual directory for the target objects that contain the data.

useTableNameAsTopicName

Targets: Apache Kafka

Indicates whether Data Ingestion and Replication writes messages that contain source data to separate topics, one for each source table or object, or writes all messages to a single topic. Valid values are:

- true. Write messages to separate table-specific topics.
- false. All messages are written to the single topic which name is specified in the topicName parameter.

includeSchemaName

Targets: Apache Kafka

When you set useTableNameAsTopicName to true, this setting adds the source schema name in the table-specific topic names. The topic names then have the format *schemaname_tablename*.

Valid values are:

- true. Add the source schema name in the table-specific topic names.
- false. Do not add the source schema name in the table-specific topic names.

tablePrefix

Targets: Apache Kafka

When you set useTableNameAsTopicName to true, this parameter specifies an optional prefix to add to the table-specific topic names. For example, if you specify myprefix_, the topic names have the format myprefix_*tablename*. If you omit the underscore (_) after the prefix, the prefix is prepended to the table name.

tableSuffix

Targets: Apache Kafka

When you set useTableNameAsTopicName to true, this parameter specifies an optional suffix to add to the table-specific topic names. For example, if you specify _mysuffix, the topic names have the format *tablename*_mysuffix. If you omit the underscore (_) before the suffix, the suffix is appended to the table name.

topicName

Targets: Apache Kafka

The name of the single Kafka topic to which all messages that contain source data will be written. Use this parameter if useTableNameAsTopicName is set to false.

stage

Targets: Snowflake

The name of internal staging area that holds the data read from the source before the data is written to the target tables. This name must not include spaces. If the staging area does not exist, it will be automatically created.

outputFormat

Targets: Amazon S3, Flat file, Google Cloud Storage, Microsoft Azure Data Lake Storage, Kafka

The format of the output file. Valid values are:

- AVRO
- CSV
- PARQUET

The default value is CSV.

Note: Output files in CSV format use double-quotation marks ("") as the delimiter for each field.

parquetFormat

Targets: Amazon S3, Flat file, Google Cloud Storage, Microsoft Azure Data Lake Storage, Kafka

If AVRO is specified as the output format in the outputFormat parameter, set this parameter to true to write data in uncompressed Parquet format. Alternatively, you can just set the outputFormat parameter to PARQUET and not include this parameter.

Important: If you set this option to true, you must install Visual C++ Redistributable Packages for Visual Studio 2013 on the computer where the Secure Agent runs.

Valid values are:

- true. Write data in uncompressed Parquet format.
- false. Write data in AVRO format.

avroFormat

Targets: Amazon S3, Flat file, Google Cloud Storage, Microsoft Azure Data Lake Storage, Kafka

If AVRO is specified as the output format, specify the format of the Avro schema that will be created for each source table or object. Valid values are:

- Avro-Flat. This Avro schema format lists all Avro fields in one record.
- Avro-Generic. This Avro schema format lists all columns or fields from a source table or object in a single array of Avro fields.
- Avro-Nested. This Avro schema format organizes each type of information in a separate record.

The default value is Avro-Flat.

avroSerializationFormat

Targets: Amazon S3, Flat file, Google Cloud Storage, Microsoft Azure Data Lake Storage, Kafka

If AVRO is specified as the output format, specify the serialization format of the Avro output file. Valid values are:

- Binary
- JSON

The default value is Binary.

avroSchemaDirectory

Targets: Amazon S3, Flat file, Google Cloud Storage, Microsoft Azure Data Lake Storage, Kafka

If AVRO is specified as the output format, specifies the local directory where Data Ingestion and Replication stores Avro schema definitions for each source table or object. Schema definition files have the following naming pattern: *schemaname_tablename*.txt

Note: If this directory is not specified, no Avro schema definition file is produced.

fixedDirectoryForEachTable

Targets: Amazon S3, Google Cloud Storage, or Microsoft Azure Data Lake Storage Gen2

For initial load tasks, indicates whether to use the source table or object names as the names of the directories to which Data Ingestion and Replication writes flat files that contain source data, for all job runs.

Valid values are:

- true. Use the source table or object names as the names of the directories for each job run.
- false. Create a new set of directories for each job run by using the following naming pattern: *tablename_timestamp*.

fileCompressionType

Targets: Amazon S3, Flat file, Google Cloud Storage, Microsoft Azure Data Lake Storage

If the output format is set to AVRO or CSV format in the outputFormat parameter, optionally specifies a file compression type. Valid values are:

- Deflate
- gzip
- Snappy

Note: If you do not specify this parameter, the output files are not compressed.

avroCompressionType

Targets: Amazon S3, Flat file, Google Cloud Storage, Kafka, Microsoft Azure Data Lake Storage

If the output format s set to AVRO in the outputFormat parameter, optionally specifies an Avro compression type. Valid values are:

- None
- Bzip2
- Deflate
- Snappy

The default value is **None**, which means no compression is used.

parquetCompressionType

Targets: Amazon S3, Flat file, Google Cloud Storage, Kafka, Microsoft Azure Data Lake Storage

If the outputFormat parameter is set to PARQUET, or if the outputFormat parameter is set to AVRO and the parquetFormat parameter is set to true, optionally specifies a Parquet compression type. Valid values are:

- None
- gzip
- Snappy

deflateCompressionLevel

Targets: Amazon S3, Flat file, Google Cloud Storage, Kafka, Microsoft Azure Data Lake Storage

If Deflate is specified in the avroCompressionType parameter, specifies a compression level from 0 to 9. The default value is 0.

addDirectoryTags

Targets: Amazon S3, Google Cloud Storage, Microsoft Azure Data Lake Storage

Indicates whether Data Ingestion and Replication adds the "dt=" prefix to the names of apply cycle directories to be compatible with the naming convention for Hive partitioning. Valid values are:

- true. Add the "dt=" prefix to the names of apply cycle directories.
- false. Do not change the names of apply cycle directories.

The default value is false.

directoryTags

Targets: Amazon S3, Google Cloud Storage, Microsoft Azure Data Lake Storage

This field is dependent on the addDirectoryTags field.

Example:

```
addDirectoryTags: true
directoryTags: \\directory
```

renamingRules

Targets: All

Optional rules for renaming the target tables that correspond to the selected source tables or objects.

Example:

```
- source: "*"
  target: "*_1"
```

dataTypeRules

Targets: Amazon Redshift, Databricks, Google BigQuery, Oracle, Snowflake, Synapse Analytics

Optional data type mapping rules to override the default mappings of source data types to target data types. The default mappings are described under Default Data Type Mappings.

Example:

```
- source: Int
  target: String
```

cdcCompatibleFormat

Targets: Amazon S3, Flat file, Google Cloud Storage, Kafka, Microsoft Azure Data Lake Storage

Indicates whether Data Ingestion and Replication includes UNDO data in the output that a job writes to the target. The default value is false.

Advanced parameters

addOperationType

Targets: Amazon S3, Apache Kafka, Flat File, Google Cloud Storage, Microsoft Azure Data Lake Storage

For incremental load and combined initial and incremental load tasks, set this parameter to true to add a metadata column that includes the source SQL operation type in the output that the job propagates to the target. For incremental load and combined initial and incremental load tasks, default is true. For initial load tasks, default is false.

addOperationTime

Targets: Amazon S3, Apache Kafka, Flat File, Google Cloud Storage, Microsoft Azure Data Lake Storage

For incremental load and combined initial and incremental load tasks, set this parameter to true to add a metadata column that includes the source SQL operation time in the output that the job propagates to the target. For initial loads, the job always writes the current date and time. Default is false for all load types.

addOperationOwner

Targets: Amazon S3, Apache Kafka, Flat File, Google Cloud Storage, Microsoft Azure Data Lake Storage

For incremental load and combined initial and incremental load tasks, set this parameter to true to add a metadata column that includes the owner of the source SQL operation in the output that the job propagates to the target. For initial loads, the job always writes "INFA" as the owner. Default is false. Not used for application ingestion and replication jobs.

addOperationTransactionId

Targets: Amazon S3, Apache Kafka, Flat File, Google Cloud Storage, Microsoft Azure Data Lake Storage

For incremental load and combined initial and incremental load tasks, set this parameter to true to add a metadata column that includes the source transaction ID in the output that the job propagates to the target for SQL operations. Default is false. Not used for application ingestion and replication jobs.

addBeforeImages

Targets: Amazon S3, Apache Kafka, Flat File, Google Cloud Storage, Microsoft Azure Data Lake Storage

For incremental load and combined initial and incremental load tasks, set this parameter to true to include UNDO data in the output that an incremental load job writes to the target. For initial loads, the job writes nulls. Default is false.

asyncWrite

Targets: Kafka

Controls whether to use synchronous delivery of messages to Kafka. Valid values are:

- true. Use asynchronous delivery. Data Ingestion and Replication sends messages as soon as possible, without regard for the order in which the changes were retrieved from the source.
- false. Use synchronous delivery. Kafka must acknowledge each message as received before Data Ingestion and Replication sends the next message. In this mode, Kafka is unlikely to receive duplicate messages. However, performance might be slower.

The default value is true.

producerConfigurationProperties

Targets: Kafka

A comma-separated list of *key=value* pairs to enter Kafka producer properties for Apache Kafka, Confluent Kafka, or Kafka-enabled Event Hubs targets.

If you have a Confluent target that uses Confluent Schema Registry to store schemas, you must specify the following properties:

```
schema.registry.url=url,
key.serializer=org.apache.kafka.common.serialization.StringSerializer,
value.serializer=io.confluent.kafka.serializers.KafkaAvroSerializer
```

You can specify Kafka producer properties in either this parameter or in the **Additional Connection Properties** field in the Kafka connection.

If you specify the producer properties in this parameter, the properties pertain to the ingestion and replication jobs associated with this task only. If you enter the producer properties for the connection, the properties pertain to jobs for all tasks that use the connection definition, unless you override the connection-level properties for specific tasks by also specifying properties in the **Producer Configuration Properties** field.

For information about Kafka producer properties, see the Apache Kafka documentation or Confluent Kafka documentation.

Custom parameters

customProperties

Targets: All

Custom properties that Informatica provides to meet special requirements. Specify these properties only at the direction of Informatica Global Customer Support.

Example:

```
customProperties:
  # cdc, s3, add target directory checkbox
  rat.cdcfile.target.tag.directories: true
  # only if CSV format selected as Output Format
  # adlsgen2, azureDLS, flat file, google, s3, kafka
  formatEncoderPrintHeader: false
  # adlsgen2, azureDLS, flat file, google, s3, kafka
  rat.audit.columns.optype: true
  rat.audit.columns.optime: true
  rat.audit.columns.opowner: true
  rat.audit.columns.optxid: true
  # kafka, actually located on step4
  checkpointAllRows: true
  checkpointEveryCommit: false
  checkpointRowCount: 0
  checkpointFrequencySecs: 0
  # any other values
  propertyName: propertyValue
```

RuntimeOptions section parameters

The runtimeOptions section of the YAML input file configures runtime options for jobs of any load type.

schemaDriftOptions

Sources: Microsoft SQL Server or Oracle, incremental load tasks and combined initial and incremental load tasks. Synapse target support ignores the value for the rename column DDL operation. Microsoft SQL Server, Oracle or PostgreSQL, incremental load tasks and combined initial and incremental load tasks are also supported.

The schema drift option to use for each supported type of DDL operation:

- addColumn
- modifyColumn
- dropColumn

- renameColumn

Valid values are:

- ignore
- replicate
- stop job
- stop table

Example:

```
schemaDriftOptions:
  addColumn: ignore
  modifyColumn: replicate
  dropColumn: replicate
  renameColumn: replicate
```

numberOfRowsInOutputFile

Targets: Amazon Redishift, Amazon S3, Google Big Query, Google Cloud Storage, Microsoft Azure Data Lake Storage, Microsoft Azure Synapse Analytics, Oracle and Snowflake

Specifies the maximum number of rows that the application ingestion and replication or database ingestion and replication task writes to an output data file on a target.

fileExtensionBasedOnFileType

Targets: Flat File, Amazon S3, Google Cloud Storage, and Microsoft Azure Data Lake Storage

Indicates whether you want the output data files to have the .dat extension.

- true. The output files have file-name extensions based on their file types.
- false. The output files have the .dat extension.

The default value is true.

Advanced parameters

applyCycleChangeLimit

Targets: Amazon S3, Google Cloud Storage, or Microsoft Azure Data Lake Storage Gen2

For application ingestion or database ingestion and replication incremental load jobs that have an Amazon S3 or Microsoft Azure Data Lake Storage Gen2 target, the number of records that must be processed before the job ends an apply cycle. When this record limit is reached, the job ends the apply cycle and writes the change data to the target. Default is 10000.

Note: Either the applyCycleChangeLimit parameter or an applyCycleInterval*time_unit* parameter must have a non-zero value.

applyCycleIntervalDays

Targets: Amazon S3, Google Cloud Storage, or Microsoft Azure Data Lake Storage Gen2

For application ingestion and replication or database ingestion and replication incremental load jobs that have an Amazon S3 or Microsoft Azure Data Lake Storage Gen2 target, the number of days that must elapse before an application ingestion and replication or database ingestion and replication job ends an apply cycle. Default is 0. You can specify this parameter with applyCycleIntervalHours, applyCycleIntervalMins, and applyCycleIntervalSecs or specify a subset of these parameters.

applyCycleIntervalHours

Targets: Amazon S3, Google Cloud Storage, or Microsoft Azure Data Lake Storage Gen2

For application ingestion and replication or database ingestion and replication incremental load jobs that have an Amazon S3 or Microsoft Azure Data Lake Storage Gen2 target, the number of hours that must elapse before an application ingestion and replication or database ingestion and replication job ends an apply cycle. Default is 0.

applyCycleIntervalMins

Targets: Amazon S3, Google Cloud Storage, or Microsoft Azure Data Lake Storage Gen2

For application ingestion and replication or database ingestion and replication incremental load jobs that have an Amazon S3 or Microsoft Azure Data Lake Storage Gen2 target, the number of minutes that must elapse before an application ingestion and replication or database ingestion and replication job ends an apply cycle. Default is 15.

applyCycleIntervalSecs

Targets: Amazon S3, Google Cloud Storage, or Microsoft Azure Data Lake Storage Gen2

For application ingestion and replication or database ingestion and replication incremental load jobs that have an Amazon S3 or Microsoft Azure Data Lake Storage Gen2 target, the number of seconds that must elapse before an application ingestion and replication or database ingestion and replication job ends an apply cycle.Default is 0.

lowActivityFlushHours

Targets: Amazon S3, Google Cloud Storage, or Microsoft Azure Data Lake Storage Gen2

The amount of time, in hours, that must elapse during a period of no change activity on the source before an application ingestion and replication or database ingestion and replication job ends an apply cycle. You can use this parameter in conjunction with lowActivityFlushMins or specify only one of them. When this time limit is reached, the ingestion job ends the apply cycle and writes the change data to the target. Default is 0.

If you do not specify a value for lowActivityFlushHours or lowActivityFlushMins, the application ingestion and replication job ends apply cycles only after either the **applyCycleChangeLimit** or cycle interval time limit is reached.

lowActivityFlushMins

Targets: Amazon S3, Google Cloud Storage, or Microsoft Azure Data Lake Storage Gen2

The amount of time, in minutes, that must elapse during a period of no change activity on the source before an application ingestion and replication or database ingestion and replication job ends an apply cycle. You can use this parameter in conjunction with lowActivityFlushHours or specify only one of them. When the overall time limit is reached, the ingestion job ends the apply cycle and writes the change data to the target. Default is 0.

If you do not specify a value for lowActivityFlushHours or lowActivityFlushMins, the application ingestion and replication or database ingestion and replication job ends apply cycles only after either the **applyCycleChangeLimit** or cycle interval time limit is reached.

Checkpoint parameters

checkpointAllRows

Targets: Apache Kafka

For application ingestion and database ingestion and replication incremental load jobs that have an Apache Kafka target, indicates whether an ingestion job performs checkpoint processing for every message that is sent to the Kafka target. When this parameter is set to true, the checkpointEveryCommit, checkpointRowCount, and checkpointFrequencySecs parameters are ignored. Default is true.

checkpointEveryCommit

Targets: Apache Kafka

For application ingestion and replication and database ingestion and replication incremental load jobs that have an Apache Kafka target, Indicates whether an ingestion job performs checkpoint processing for every commit that occurs on the source. Default is false. If you set this parameter to true, do not also specify checkpointAllRows.

checkpointRowCount

Targets: Apache Kafka

For application ingestion and replication and database ingestion and replication incremental load jobs that have an Apache Kafka target, the maximum number of messages that a job sends to the target before adding a checkpoint. If you set this option to 0, an ingestion and replication job does not perform checkpoint processing based on the number of messages. If you set this option to 1, an ingestion job adds a checkpoint for each message. Default is 0. If you set this parameter to a non-zero number, do not also specify checkpointAllRows.

checkpointFrequencySecs

Targets: Apache Kafka

For application ingestion and replication and database ingestion and replication incremental load jobs that have an Apache Kafka target, the maximum number of seconds that must elapse before an ingestion job adds a checkpoint. If you set this option to 0, an ingestion and replication job does not perform checkpoint processing based on elapsed time. Default is 0. If you set this parameter to a non-zero number, do not also specify checkpointAllRows.

Custom parameters

customProperties

Custom properties that Informatica provides to meet special requirements. Specify these properties only at the direction of Informatica Global Customer Support.

Example:

```
customProperties:
  rat.cdcfile.cycle.dml.limit: 10000
  rat.cdcfile.cycle.interval.days: 0
  rat.cdcfile.cycle.interval.hours: 0
  rat.cdcfile.cycle.interval.mins: 0
  rat.cdcfile.cycle.interval.secs: 0
  rat.cdcfile.lowact.flush.hours: 0
  rat.cdcfile.lowact.flush.mins: 0
```

Schedule section parameters

The schedule section of the YAML input file consists of a single parameter that specifies the task schedule. A large number of scheduled jobs can lead to increased resource consumption.

schedule

The name of a predefined schedule that you want to use for running run job instances for an initial load task instead of manually starting the job after it is deployed from one of the monitoring interfaces.

Default values

The defaults.yml file is in the /ymls folder. This file contains the default values for location, salesforceAPI, selectionRules, cdcIntervalMins, fetchSize, and restartPointForIncrementalLoad. You can customize this file according to your needs.

For example, if you did not specify selectionRules in the yml file with your task configuration values that you customized, the values from the defaults.yml file will be applied.

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